Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.31666	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.4171	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1500.70588	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.171366	0.171366	1616.67754	0.68734	13.43

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.31666.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.4171.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1500.70588.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.178906.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.171366.
- Held-out runtime ms: 1616.67754.
- Robustness dispersion across held-out structure families: 0.012712.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.280276, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.240128, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.135099, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.108363, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.092963, errors=[]

Judge Note

Phase-9 CVRP Solver-Evolution Suite Summary

Metric	Nearest Neighbor	Clarke-Wright Savings	Regret Insertion LS	Solver Evolution (Evolved)
Feasibility (heldout)	1.0	1.0	1.0	1.0
Heldout Feasible Gap	0.2734	**0.0738 (best)**	0.4664	0.1714
Heldout Runtime (ms)	42.3	77.4	1500.7	1616.7
Robustness Dispersion	0.0028	0.0078	0.0869	0.0127
Code Novelty / Complexity	0 / 0	0 / 0	0 / 0	0.6873 / 13.43

Analysis

- **Feasibility:** All solvers achieved perfect feasibility (1.0) on held-out data.
- **Objective gap:** The Clarke-Wright Savings baseline achieved the lowest (best) heldout gap (0.0738), outperforming the evolved solver (0.1714) by a substantial margin.
- **Runtime:** The evolved solver and the Regret Insertion local search are much slower (≈1.6 seconds) compared to simpler baselines (nearest neighbor: 42 ms, Clarke-Wright: 77 ms).
- **Robustness:** The evolved solver shows moderate robustness dispersion (0.0127), better than Regret Insertion but worse than the Clark-Wright baseline.
- **Novelty/complexity:** The solver evolution introduced significantly more complexity and novelty in code compared to baselines (0 novelty, 0 complexity).

Conclusion

The evolved solver maintained feasibility and showed reasonable robustness, but **did not clearly beat the fixed baselines** in objective gap or runtime. Specifically, the Clarke-Wright Savings baseline outperformed the evolved solver in both solution quality and computational efficiency on held-out CVRP instances across multiple structure families.